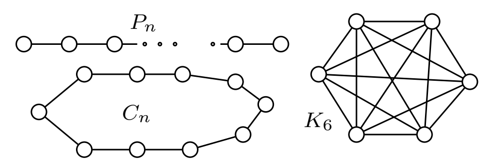
FCS Practice Session 1.

September 7, 2026

Summary

Definition $G = (V, E)$ is a *simple graph*, if (1) $V \neq \emptyset$ and (2) $E \subseteq \binom{V}{2} := \{\{u, v\} : u, v \in V, u \neq v\}$
$V(G)$ denotes the set of *vertices (nodes)* of G, while $E(G)$ denotes the *edges* of G, that is $G = (V(G), E(G))$ for a graph G. The graph $G = (V, E)$ is *finite*, if V and E are both finite sets.
Definition The *diagram* of graph G is a drawing of G where the vertices correspond to points (of the plane) and the edges correspond to not self-intersecting curves connecting the end the two endvertices of the edges.
Definition Edge $e = \{u, v\}$ is denoted by $e = uv$; u and v are the *endvertices* of edge e. Vertices u and v are *adjacent*, if e is an edge of the graph. If $e = \{u, v\}$ then e and u, resp., v, are *incident*. Edges e, f are *parallel* if their endvertices are the same. A *loop* is an edge whose two endvertices are identical. Graph $G = (V, E)$ is *simple* if it does not contain parallel edges or loops.
Definition The *degree* $d(v)$ of a vertex v of graph G is the number of edges incident with v (loops count twice) : $d(v) := |\{e \in E : v \text{ is an endvertex of } e\}| + |\{e \in E : e \text{ loop incident with } v\}|$



Proposition (HSL) If G is a finite graph, then the sum of degrees of its vertices is $2|E(G)|$.
K_n is the n-vertex *complete graph*: any two its vertices are adjacent.
Definition P_n is *n-vertex path*, C_n az *n-vertex cycle* (see the picture.)
G is *(k)-regular*, if all its degrees are identical ($= k$). $\Delta(G)$ (resp. $\delta(G)$) is the max, (resp. min) degree of G.
Definition The *complement* of the simple graph G is the graph $\overline{G} := (V(G), \binom{V}{2} \setminus E(G))$. (Two vertices are adjacent in $\overline{G}$ if and only if they are not adjacent in G.) Finite graphs G_1 and G_2 are *isomorphic* ($G_1 \cong G_2$), if vertices of both G_1 and G_2 can be numbered from 1 and n, so for arbitrary i, j the number of edges between i and j in G_1 is the same as in G_2.
In other words, G_1 and G_2 are isomorphic ($G_1 \cong G_2$), if there exists bijections $\phi_V : V(G_1) \to V(G_2)$ and $\phi_E : E(G_1) \to E(G_2)$ such that $uv \in E(G) \iff \phi_V(u)\phi_V(v) = \phi_E(uv)$.

Definition A *walk* in graph G is a sequence $(v_1, e_1, v_2, e_2, \ldots, v_k)$, where $e_i = v_i v_{i+1} \in E(G)$ $(\forall i)$ and $e_1, e_2, \ldots, e_{k-1}$ are pairwise distinct. A walk is *closed*, if $v_1 = v_k$.
Definition A *path* (resp. *cycle*) such a (closed) walk, whose vertices are pairwise distinct (ecept for the start and end vertex). In a simple graph a path or cycle can be identified with its sequence of vertices.
Proposition In a graph G there exists a walk between u and v, iff there exists a path between u and v.
Definition Graph G is *connected*, if there exists a walk between any two its vertices.
K is a *component* of G if between any $u, v \in K$ there exists walk in G, but there exists no uv-walk if $u \in K$, $v \in V(G) \setminus K$. **Proposition** Every graph can be decomposed into components, and this decomposition is unique.
Proposition Edge Addition Lemma (EAL): Exactly one of the following is true for graph $G + e$: (1) There exists no cycle through e, and $G + e$ has one less component than G, (2) there exists a cycle through e, and $G + e$ has the same number of components as G has.
Definition Let $G = (V, E)$ be a graph, $e \in E$, $v \in V$. Then $G - e = (V, E \setminus \{e\})$ is the result of an *edge deletion*; the graph $G - v$ obtained by *vertex deletion* is $G - v = (V \setminus \{v\}, E')$, where E' is obtained from E by deleting all edges incident with v.
Definition Graph H is a *induced/spanning/without adjective* subgraph of G, if H is obtained from G by vertex deletions/edge deletions/vertex and edge deletions.
Proposition H is a (1) subgraph (2) spanning subgraph (3) induced subgraph of G, iff (1) $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$, (2) $V(H) = V(G)$ and $E(H) \subseteq E(G)$ resp. (3) $V(H) \subseteq V(G)$ and $E(H)$ consists of those edges from $E(G)$ whose both endvertices are in $V(H)$.
G has one less edges than vertices.

Problems

1. Place two white and two dark knights in the corners of a 3×3 chessboard so that the identically colored knights are on diametrically opposite corners. The knights can move by the usual way known from chess, so that two knights xcannot stand on the same square at the same time. Is it possible to move the knights so that they stand in the corners at the end, but the diametrically opposite ones are of different colors?

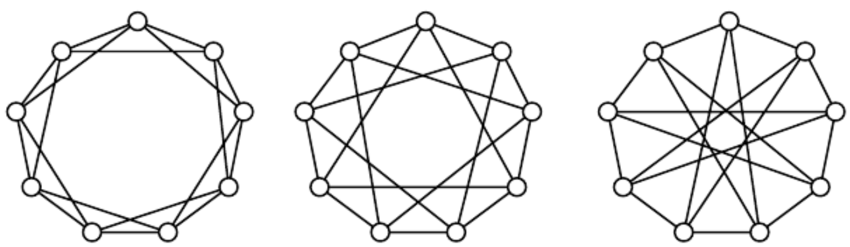
2. Are there simple graphs with degree sequence a.) 1, 2, 2, 3, 3, 3, resp. b.) 1, 1, 2, 2, 3, 4, 4?

3. Determine 2-regular graphs. What are those graph that satisfy $\Delta(G) \leq 2$?

4. Show that the number of odd-degree vertices of a finite graph is even. Show that this is not necessarily true for infinite graphs

5. Prove that if the degree of vertex $u \in V(G)$ is odd, then there exists such a uv-path in G that $u \neq v$ and the degree of v is also odd.

6. Determine which of the following three graphs are isomorphic to each other.



7. Let G be a simple graph and assume that $d(v) \geq \frac{n}{2}$ holds for all of its vertices. Prove that G is connected.

8. Let us assume that the outdegree of every vertex of a directed graph is 22. It is also known, that the indegree of each vertex, except for three special vertices, is 22, furthermore the indegree of two of the three exceptional vertices is 12. What is the indegree if the third exceptional vertex?

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9. Determine all finite simple graphs that do not have two vertices of the same degree.

10. Prove that if the degree sequence of a 6-vertex graph is 2, 2, 2, 4, 5, 5, then it is not simple.

11. Show that if graph $G = (V, E)$ satisfies $|V| = 11$ and $|E| = 45$, then it has a vertex of degree at least 9.

12. Alice and Bob plays the following game. Alternatingly they draw edges between n given vertices. The player who first obtaines a cycle looses the game. Who wins if both players play optimal strategy, the one who moves first, or the other one?

13. Let us assume that simple graph G has 100 vertices and each has degree at least 33, furthermore, there is vertex of G that is incident with at least 66 edges. Prove that G is connected. (Midterm '15)

14. How many edges does the 66-vertex graph G have that satisfies the following two properties?
    (a) At least 10 edges must be deleted from G in order to obtain a cycle-free graph.
    (b) At least 33 edges must be added to G in order to obtain a connected graph.

15. Find (up to isomorphism) all n-vertex m-edge graphs for
    a) $n = 5, m = 2$ b) $n = 5, m = 8$ c) $n = 5, m = 3$.

16. Are there 5-vertex graphs that are isomorphic to their complements? How about 6 vertex graphs?

17. Let G be an arbitrary n-vertex graph. Prove that if G is connected, then it has at least $n - 1$ edges. Also show, that if it has n edges, then it contains a cycle.

18. Is it possible to obtain a connected graph form a graph of 88 vertices, 42 components and 111 edges in sequence of steps, where one step consists of adding one edge and deleting two edges at the same time? (An edge deleted earlier can be added later.)

19. Is it possible to obtain a cycle-freee simple graph from a 77-vertex simple graph of 66 edges and 42 components by applying a sequence of steps where a step consists of deletion of one edge and addition of two edges? (An edge deleted earlier is possible to be added back later.)

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20. Simple graph G has $2k$ vertices. Every vertex has degree at least $k-1$, and there exists at least one vertex of degree k. Show that G is connected.

$\boxed{\star}$ At most how many vertices may a graph G have, that has 123 edges and its edges can be colored using colors red, white and green (every edge gets exactly one color), so that between any two vertices of G there exists paths consisting of only red, only white as well as only green edges?

21. Prove that the edges of a simple graph $G = (V, E)$ can be oriented so that the directed graph obtained does not contain directed cycle.

22. Let e, f and g be edges of simple, connected graph G. Assume that G remains connected if any one of its edges is deleted, however neither $G - e - g$ nor $G - f - g$ is connected. Prove that graph $G - f - g$ is also disconnected.

23. All rows of an $n \times n$ table T are pairwise distinct. Prove that T has a column such that table obtained by the deletion of that columns still has pairwise distinct rows.

24. Is it possible to color the edges of the complete graph K_4 using colors red, white and green in such a way, that however way two of the colors are chosen of the three, the two graphs formed by the edges of these two colors, respectively, are isomorphic to each other?

25. Prove that among any 13 people there exists one who knows at least 6 others, or there exists 3 people who are pairwise unknown to each other. (Acquaintance is symmetric.)

26. Show that if a rectangle T is tiled by rectangles so that each tile has at least one side of integer length, then T also has a side of integer length.